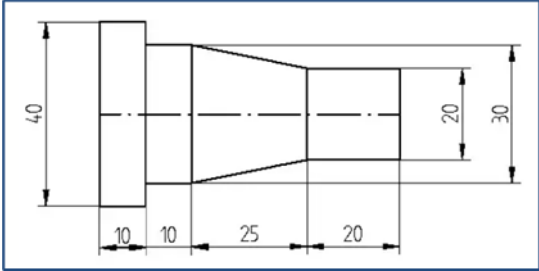


Course Code: B23ME4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B. Tech. I Semester MODEL QUESTION PAPER					
CAD/CAM					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define CAD and CAM.	1	2	2
	b).	List any two benefits of CAD.	1	2	2
	c).	What is a geometric transformation?	2	2	2
	d).	Define wireframe modeling.	2	2	2
	e).	What is CNC?	3	2	2
	f).	Write the advantages of CNC machines.	3	2	2
	g).	What is group technology?	4	2	2
	h).	Define FMS.	4	2	2
	i).	What is CIM?	5	2	2
	j).	Define process planning.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2		Explain the applications of computers in design	1	3	10
		OR			
3.		Illustrate the 3D geometric transformations with suitable examples	1	3	10
		UNIT-2			
4.		Differentiate between wireframe, surface and solid modeling.	2	3	10
		OR			
5.		Explain Bezier curves and cubic spline curves.	2	3	10
		UNIT-3			
6.		Explain the structure and components of a CNC machine.	3	3	10
		OR			
7		Write the manual part programming for the following component.	3	3	10

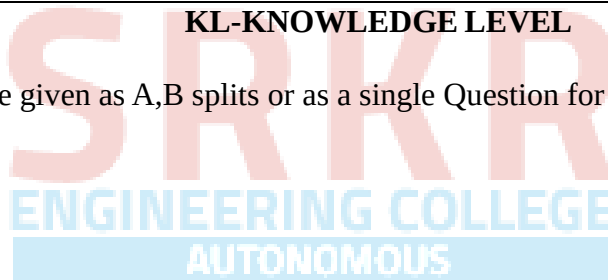
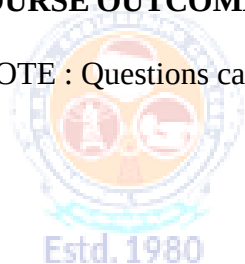
					
		UNIT-4			
8.		Explain the implementation of OPTIZ coding systems	4	3	10
		OR			
9.		Illustrate the types of layouts in FMS.	4	3	10
		UNIT-5			
10.		Explain Computer Integrated Manufacturing (CIM) and its benefits.	5	3	10
		OR			
11.		Explain various approaches to computer-aided process planning.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23ME4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Artificial Intelligence.	1	1	2
	b).	Define Rational Agent.	1	1	2
	c).	Define Machine Learning.	2	1	2
	d).	What is overfitting?	2	1	2
	e).	Define clustering.	3	1	2
	f).	Define K-Nearest Neighbour (KNN).	3	1	2
	g).	What is a perceptron?	4	1	2
	h).	What is Convolutional Neural Network (CNN)?	4	1	2
	i).	What is ensemble learning?	5	1	2
	j).	What is Random Forest?	5	1	2
5 x 10 = 50 Marks					
		UNIT-I	CO	KL	M
2.		Explain the structure of intelligent agents with a neat diagram.	1	2	10
		OR			
3.		Explain Smart Manufacturing and Digital Twins with suitable industrial applications.	1	3	10
		UNIT-II			
4.		Explain the differences between supervised and unsupervised learning.	2	2	10
		OR			
5.		Illustrate the working of Decision Tree algorithms.	2	3	10
		UNIT-III			
6.		Illustrate the K-Means clustering algorithm with steps.	3	3	10
		OR			
7.		Explain the applications of Computer Vision in manufacturing and defect detection.	3	3	10
		UNIT-IV			
8.		Illustrate the architecture of Artificial Neural Networks.	4	3	10
		OR			
9.		Discuss the working of Genetic Algorithms and their operators	4	2	10
		UNIT-V			

10.		Explain ensemble learning methods such as Bagging, Boosting, and Random Forest.	5	2	10
		OR			
11.		Illustrate methods for evaluating machine learning algorithms.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23ME4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ADVANCED MATERIALS					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are super alloys and mention their key applications?	1	1	2
	b).	Define metallic foams and state their advantages.	1	1	2
	c).	Define sintering in ceramic processing.	2	1	2
	d).	What are cermets? Give one application.	2	1	2
	e).	What are polymer matrix composites (PMCs)?	3	1	2
	f).	What is the role of reinforcements in composite materials?	3	1	2
	g).	List the types of shape memory alloys (SMAs).	4	1	2
	h).	Mention any two applications of FGMs.	4	1	2
	i).	Mention any two properties of materials at nanoscale.	5	1	2
	j).	List the types of nanostructures.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Discuss the characteristics and applications of titanium and nickel-based alloys in aerospace engineering.	1	3	10
		OR			
3.		Describe the selection and evaluation of materials for extreme environments, including cryogenic and space applications.	1	3	10
		UNIT-2			
4.		Explain the classification, characteristics, and applications of ceramics in detail.	2	3	10
		OR			
5.		Discuss the properties and applications of Al ₂ O ₃ and Si ₃ N ₄ ceramics.	2	3	10
		UNIT-3			
6.		Describe the structure, properties, and applications of fiber-reinforced composites.	3	3	10
		OR			

7.		Explain the classification of composite materials, including polymer, metal, ceramic, and carbon-carbon composites with applications.	3	3	10
		UNIT-4			
8.		Discuss the composition and characteristics of SMAs, highlighting their engineering applications.	4	3	10
		OR			
9.		Explain the properties and applications of FGMs, emphasizing their advantages over conventional materials.	4	3	10
		UNIT-5			
10.		Explain the properties of nanomaterials at nanoscale and how they differ from bulk materials.	5	3	10
		OR			
11.		Describe different types of nanostructures such as nanowires, nanotubes, and nanocomposites with applications.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23ME4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SMART MANUFACTURING					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define circular manufacturing.	1	1	2
	b).	Differentiate Industry 4.0 and Industry 5.0.	1	2	2
	c).	Identify a smart sensor used for tool wear monitoring.	2	1	2
	d).	Draw the block diagram of a smart sensor interface used in IIoT.	2	1	2
	e).	Explain the Cognition level in 5C architecture.	3	2	2
	f).	Compare SCADA systems and Cyber-Physical Systems.	3	2	2
	g).	Explain the role of a Digital Twin in manufacturing.	4	2	2
	h).	Classify the Six Big Losses in manufacturing.	4	2	2
	i).	Explain the TCP/IP model used in industrial communication.	5	2	2
	j).	Compare MQTT and HTTP protocols.	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the development from Industry 3.0 to Industry 4.0. How do smart machines support mass customization?	1	3	5
	b).	Explain what are the challenges do small industries face while implementing smart manufacturing? Suggest two solutions.	1	3	5
OR					
3.	a).	Explain how circular economy concepts can be used in an automobile assembly plant. Mention three stages of resource optimization.	1	3	5
	b).	Explain the stages of Industry 4.0 evolution. At which stage does a factory move from visibility to predictability? Give an example.	1	3	5
UNIT-2					
4.	a).	A factory faces downtime due to motor overheating. Design a sensor system to monitor temperature and vibration.	2	3	5
	b).	Explain the role of smart actuators in a smart manufacturing system. How are they different from traditional actuators?	2	3	5
OR					
5.	a).	Explain how old machines can be connected to an IIoT network using	2	3	5

		smart sensors.			
	b).	Draw and explain the block diagram of an IIoT architecture from sensor to cloud.	2	3	5
		UNIT-3			
6.	a).	Explain the 5C architecture for a robotic welding system.	3	3	5
	b).	Compare traditional programmed manufacturing systems with Cyber-Physical autonomous systems.	3	3	5
		OR			
7.	a).	Explain how the Cyber level in 5C architecture uses Digital Twins for production planning.	3	3	5
	b).	Explain how Cyber-Physical Systems help in reducing the Six Big Losses.	3	3	5
		UNIT-4			
8.	a).	Explain the steps involved in creating a Digital Twin for a wind turbine to predict blade damage.	4	3	5
	b).	Differentiate Supervised and Unsupervised Machine Learning. Explain one method used for anomaly detection in conveyor systems.	4	3	5
		OR			
9.	a).	Develop a predictive maintenance method for a hydraulic press using AI/ML. Which Six Big Loss is reduced by this method?	4	3	5
	b).	Explain the concept of Multiverse in Digital Twins with AR/VR for remote maintenance.	4	3	5
		UNIT-5			
10.	a).	Explain the requirements of a mesh network in a smart warehouse. Why is it better than star topology?	5	3	5
	b).	Explain the use of MQTT or CoAP protocols when many sensors send data with low congestion.	5	3	5
		OR			
11.	a).	Explain the importance of Physical Layer IoT protocols in high-speed motion control systems.	5	3	5
	b).	Draw the TCP/IP model and explain how safe transmission of an Emergency Stop signal is achieved in a factory network.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23ME4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
MICRO ELECTRO MECHANICAL SYSTEMS					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define MEMS and mention any two applications.	1	3	2
	b).	What is a piezoelectric material?	1	3	2
	c).	Define Shape Memory Alloy (SMA).	2	3	2
	d).	State any two applications of thermal actuators.	2	3	2
	e).	What are MOEMS? Give one application	3	3	2
	f).	What is a micro mirror?	3	3	2
	g).	Define microfluidics.	4	3	2
	h).	Mention two uses of micro pumps.	4	3	2
	i).	Define Chem-Lab-on-Chip (CLOC).	5	3	2
	j).	Define calorimetric spectroscopy.	5	3	2
5 x 10 =50Marks					
UNIT-1					
2.	a).	Describe the working of a simple capacitive sensor.	1	3	5
	b).	Describe surface micromachining and its steps.	1	3	5
OR					
3.		Explain the LIGA process in detail with neat diagrams. Mention its advantages and applications.	1	3	10
UNIT-2					
4.	a).	Illustrate the working principle of thermocouple and thermistor sensors used in MEMS.	2	3	5
	b).	Explain the working principle of shape memory alloy (SMA) actuators	2	3	5
OR					
5.	a).	Analyze the operation of U-shaped electro thermal actuators used in MEMS.	2	3	5
	b).	Analyze the operation of Peltier effect heat pumps used in MEMS devices.	2	3	5
UNIT-3					

6.	a).	Illustrate the basic principle of Micro-Opto-Electro Mechanical Systems (MOEMS).	3	3	5
	b).	Explain the working of micro mirrors used in optical MEMS devices.	3	3	5
		OR			
7.	a).	Illustrate the working of Digital Micro Mirror Device (DMD).	3	3	5
	b).	Explain the operation of MEMS magnetic actuators.	3	3	5
		UNIT-4			
8.	a).	Explain the basic concept and applications of micro-fluidic systems.	4	3	5
	b).	Illustrate the working principle of electro-osmosis flow in micro-fluidic devices.	4	3	5
		OR			
9.	a).	Illustrate the working of RF MEMS switches used in communication systems.	4	3	5
	b).	Explain the operation of RF MEMS resonators and filters.	4	3	5
		UNIT-5			
10.	a).	Explain the basic sensing mechanism used in chemical microsensors.	5	3	5
	b).	Illustrate the working principle of chemo-resistors used in MEMS sensors.	5	3	5
		OR			
11.	a).	Explain the working of electronic nose (E-nose) used in chemical detection.	5	3	5
	b).	Explain the working of mass sensitive chemical sensors used in MEMS.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23ME4106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
FUNCTIONAL MATERIALS AND APPLICATIONS					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is Miller–Bravais indices?	1	2	2
	b).	What is elastic strain energy?	1	2	2
	c).	What is classical free electron theory?	2	2	2
	d).	What is a quantum interference device?	2	2	2
	e).	List the examples for direct and indirect band gap semiconductors.	3	2	2
	f).	What is the Hall effect?	3	2	2
	g).	What is optical emission?	4	2	2
	h).	What is a solar cell?	4	2	2
	i).	What are Zener–Bloch oscillations?	5	2	2
	j).	What is resonant tunneling?	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain about coordination number and atomic packing factor for BCC structure	1	3	5
	b).	Define planar density and explain its significance in crystal structures.	1	3	5
OR					
3.	a).	Explain edge dislocation with a neat sketch	1	3	5
	b).	Discuss on phase changes in crystalline materials	1	3	5
UNIT-2					
4.	a).	Explain Fermi–Dirac statistics and its significance in metals	2	3	5
	b).	Explain diamagnetism, paramagnetism and ferromagnetism	2	3	5
OR					
5.		Derive the expression for thermal conductivity using classical free electron theory.	2	3	10
UNIT-3					
6.	a).	Sketch and explain the energy band diagram of semiconductors	3	3	5
	b).	Explain the variation of carrier concentration with temperature in semiconductors.	3	3	5

		OR			
7.		Sketch and explain the construction, working and characteristics of a Schottky diode.	3	3	10
		UNIT-4			
8.	a).	Explain optical absorption and emission in semiconductors.	4	3	5
	b).	Explain the excitonic state in organic semiconductors.	4	3	5
		OR			
9.	a).	Explain the construction and working of solar cells.	4	3	5
	b).	Explain about optical switching devices	4	3	5
		UNIT-5			
10.	a).	Explain the concept of quantum confinement	5	3	5
	b).	Discuss quantum interference effects	5	3	5
		OR			
11.	a).	Distinguish between active and passive optoelectronic devices	5	3	5
	b).	Explain the properties of carbon nanotubes	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

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Course Code: B23ME4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HYDROGEN AND FUEL CELL TECHNOLOGY					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the key properties of hydrogen that make it suitable as a fuel?	1	1	2
	b).	What is the main challenge in hydrogen storage?	1	1	2
	c).	Mention any two advantages of gasification for hydrogen production.	2	1	2
	d).	What is partial oxidation?	2	1	2
	e).	List out the key physical properties of hydrogen relevant to storage?	3	1	2
	f).	Define chemical hydride storage.	3	1	2
	g).	Mention the uses of hydrogen in IC engines.	4	1	2
	h).	What is a hydrogen dual-fuel engine?	4	1	2
	i).	Define the role of thermodynamics in fuel cell operation.	5	1	2
	j).	List out any few applications of microbial fuel cells.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the concept of hydrogen pathways and describe its current uses in different sectors.	1	3	5
	b).	Discuss briefly on the infrastructure requirements for hydrogen production, storage, dispensing, and utilization.	1	3	5
OR					
3.	a).	Describe the different methods of hydrogen production and their relative advantages.	1	3	5
	b).	Explain how hydrogen can contribute to future energy security and sustainability.	1	3	5
UNIT-2					
4.	a).	Explain thermo-chemical water splitting and evaluate its potential for large-scale hydrogen generation.	2	3	5
	b).	Explain nuclear thermal catalytic methods and discuss their application in hydrogen production.	2	3	5
OR					
5.	a).	Describe biological hydrogen production methods such as anaerobic	2	3	5

		digestion and fermentation using microorganisms.			
	b).	Explain the principle of electrolysis and outline the working of a PM-based electrolyser.	2	3	5
		UNIT-3			
6.	a).	Compare physical and chemical hydrogen storage methods with examples.	3	3	5
	b).	Discuss zeolite-based hydrogen storage and its advantages over conventional methods	3	3	5
		OR			
7.	a).	Explain cryogenic hydrogen storage and highlight its challenges.	3	3	5
	b).	Discuss the future prospects of advanced hydrogen storage technologies in building a hydrogen economy.	3	3	5
		UNIT-4			
8.		Elaborate the general overview of hydrogen utilization in different sectors and compare hydrogen utilization in IC engines, gas turbines, and burners.	4	3	10
		OR			
9.		Describe the application of hydrogen in power plants and the future prospects of hydrogen utilization in achieving a sustainable energy economy.	4	3	10
		UNIT-5			
10.	a).	Describe the working of a fuel cell with neat diagrams.	5	3	5
	b).	Explain the methods of performance evaluation of fuel cells.	5	3	5
		OR			
11.	a).	Compare batteries and fuel cells in terms of working and efficiency.	5	3	5
	b).	Discuss the applications of fuel cells in transportation, stationary power, and portable devices.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23ME4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
NANO TECHNOLOGY					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	State Bragg's Law and its importance in crystallography.	1	1	2
	b).	Differentiate between crystal directions and crystal planes.	1	1	2
	c).	What is surface-to-volume ratio?	2	1	2
	d).	Define the term "Band Structure" in the context of nanomaterial.	2	1	2
	e).	What is the Sol-Gel process?	3	1	2
	f).	Briefly explain the principle of Ball milling.	3	1	2
	g).	What is the primary use of XRD in nanomaterial characterization?	4	1	2
	h).	What information can be derived from Raman Spectroscopy?	4	1	2
	i).	Briefly describe the structure of Graphene.	5	1	2
	j).	What are Quantum Dots?	5	1	2
5 x 10 = 50 Marks					
		UNIT-I	CO	KL	M
2.	a)	Discuss how the materials change their behavior at nano level?	1	3	6
	b)	Illustrate crystal symmetries.	1	3	4
		OR			
3.	a)	Write a short note on classification of nanomaterials.	1	3	6
	b)	Briefly explain Anisotropy in a single crystal.	1	3	4
		UNIT-II			
4.		Discuss in detail the optical and electronic properties of metal nanoparticles	2	2	10
		OR			
5.	a)	Briefly discuss the effects of nano scale dimension on Mechanical properties.	2	3	6
	b)	What are the applications of nanomaterials in mechanical engineering? Explain.	2	3	4
		UNIT-III			
6.		Briefly explain the following: (a) Ion implantation and (b) Electron beam lithography.	3	3	10
		OR			
7.	a)	Discuss the top down and bottom-up nanofabrication methods with	3	3	6

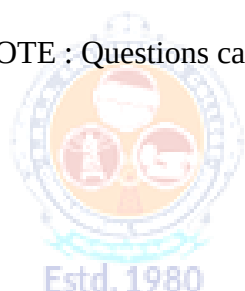
		their merits and demerits.			
	b)	Write a short note on sol-gel method and explain its principle.	3	3	4
		UNIT-IV			
8.	a)	What is the basic principle in Scanning Electron Microscope? How is it different from optical microscopy? Explain.	4	3	6
	b)	Explain how to characterize a material with scanning electron microscope (SEM) with neat sketch.	4	3	4
		OR			
9.	a)	Differentiate SEM and TEM.	4	3	6
	b)	Illustrate in detail about Raman spectroscopy.	4	3	4
		UNIT-V			
10.	a)	Discuss briefly about carbon nano tubes and their fields of application.	5	3	6
	b)	Discuss briefly about graphene.	5	3	4
		OR			
11.	a)	Discuss the applications of Nano technology in energy and environment.	5	3	6
	b)	What are the potential effects of nanomaterials to environment?	5	3	4

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23ME4110					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
CONVENTIONAL AND FUTURISTIC VEHICLE TECHNOLOGY					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Describe Gasoline Direct Injection.	1	2	2
	b).	Describe the function of a turbocharger in an engine	1	2	2
	c).	Describe the function of EMS.	2	2	2
	d).	What is the advantage of using dual-fuel combustion?	2	2	2
	e).	Why is ethanol considered a renewable fuel source?	3	2	2
	f).	What is hydrogen fuel and how is it produced?	3	2	2
	g).	Define a parallel hybrid system.	4	2	2
	h).	Name the main components of a pure electric vehicle	4	2	2
	i).	Describe the role of the electrolyte in a fuel cell.	5	2	2
	j).	What is a Solid Oxide Fuel Cell?	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain in detail the construction, working of a Gasoline Direct Injection (GDI) system.	1	2	5
	b).	Explain in detail about the types of VVT mechanisms in the IC engine.	1	2	5
OR					
3.	a).	Write short notes about the types of turbochargers	1	2	5
	b).	Illustrate with a neat sketch of the working of a supercharger	1	2	5
UNIT-2					
4.	a).	Explain in detail the stages of combustion in a Spark Ignition engine.	2	2	5
	b).	Illustrate with a suitable sketch the working principle of Digital Twin Spark Ignition System.	2	3	5
OR					
5.	a).	Explain the working principle of the Homogeneous Charge Compression Ignition (HCCI) system	2	2	10
UNIT-3					

6.	a).	Explain the properties of alcohol fuels and discuss their suitability as alternative fuels for internal combustion engines.	3	2	5
	b).	Discuss the storage, handling, and distribution issues associated with alcohol fuels in comparison to petroleum based fuels.	3	2	5
		OR			
7.	a).	Discuss briefly about types of ammonia fuels used in the internal combustion engine.	3	2	5
	b).	Explain in detail about the challenges and transportation of ammonia fuels.	3	2	5
		UNIT-4			
8.	a).	Explain the working principle of a conventional hybrid electric vehicle (HEV) with a neat block diagram.	4	2	5
	b).	Explain in detail about techniques to Enhance Hybrid Performance in Hybrid Electric Vehicles (HEVs).	4	2	5
		OR			
9.	a).	Explain with a suitable sketch about a battery electric vehicle and a plug-in hybrid electric vehicle.	4	2	5
	b).	Explain in detail about the components and working of electric vehicle in detail	4	2	5
		UNIT-5			
10.	a).	Explain with a suitable sketch about fuel cell fuel cell operating principle.	5	2	5
	b).	Explain in detail about the Proton Exchange Membrane Fuel Cell with a suitable sketch	5	2	5
		OR			
11.	a).	Describe in detail the structure and function of a fuel cell stack used in FCEVs.	5	2	5
	b).	Describe the difference between compressed and liquid hydrogen storage in FCEVs.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23ME4111					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AUTONOMOUS SYSTEMS					
Mechanical Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define an Intelligent Agent in the context of autonomous systems?	1	1	2
	b).	What is meant by the "Society of Agents"?	1	2	2
	c).	List the different Degrees of Autonomy as per standard classifications?	2	1	2
	d).	Briefly explain the role of Tele-operation in unmanned systems?	2	2	2
	e).	What is the difference between kinematics and dynamics in robot modeling?	3	4	2
	f).	Mention two specific challenges in underwater vehicle propulsion?	3	2	2
	g).	What is Sensor Fusion, and why is it critical for autonomy?	4	2	2
	h).	Define the function of an IMU (Inertial Measurement Unit)	4	2	2
	i).	What is meant by Frame Rotation in aerial robot representation?	5	2	2
	j).	Name two common Networking Protocols used in autonomous systems.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the various Agent Architectures used in autonomous systems with neat diagrams.	1	2	5
	b).	Discuss the significance of RoboCup as a case study for multi-agent systems and collaborative robotics.	1	2	5
		OR			
3.	a).	Describe the principles of Distributed Problem Solving and Planning.	1	3	5
	b).	How does collaborative robotics differ from traditional industrial robotics in terms of agent interaction?	1	2	5
		UNIT-2			
4.	a).	Elaborate on the architectural design of Automated Guided Vehicles (AGVs) and Drones.	2	4	5
	b).	Compare and contrast Reactive Systems and Real-time Systems in autonomous navigation.	2	4	5
		OR			

5.	a).	Discuss the applications of Augmented Reality (AR) and Virtual Reality (VR) in monitoring and controlling autonomous systems.	2	4	5
	b).	Explain the hierarchy of autonomy levels from manual control to full automation.	2	4	5
		UNIT-3			
6.	a).	Provide a detailed classification of Unmanned Aerial Vehicles (UAVs) based on wing types (fixed, flapping, and rotary).	3	2	5
	b).	Explain the basic forces and kinematic modeling involved in ground-wheeled vehicles.	3	2	5
		OR			
7.	a).	Describe the motion control strategies for legged robots, specifically focusing on the difference between biped and quadruped motion.	3	4	5
	b).	Discuss the control techniques used for AUVs (Autonomous Underwater Vehicles) involving fins and propulsion.	3	4	5
		UNIT-4			
8.	a).	Explain the working principles and characteristics of LIDAR and RTK-GPS in the context of autonomous navigation.	4	4	5
	b).	Describe the popular computing platforms (like NVIDIA Jetson or Raspberry Pi) used for processing high-volume sensor data.	4	4	5
		OR			
9.	a).	Compare the use of Harmonic Drives and Linear Actuators in robotic joints.	4	2	5
	b).	Discuss the process of sensor data aggregation and how thermal imaging contributes to vision-based autonomy.	4	2	5
		UNIT-5			
10.	a).	Derived from the Equations of Motion, explain the State-Space Representation for an aerial robot	5	3	5
	b).	Differentiate between PID Control and LQR (Linear Quadratic Regulator) control for flight stabilization	5	4	5
		OR			
11.	a).	Elaborate on the strategies for Collision-free Navigation and motion planning in a 3D environment.	5	3	5
	b).	Explain Vision-based Guidance for aerial robots and mention the common Networking Protocols used for communication	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks